A
Domain Specific Project Report
On
"Land Price and Tax Prediction System"

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INDEX

Sr. No	Title	Page no
1	Synopsis	1
2	Introduction	14
	2.1 Overview of the project	15
	2.2 Problem Statement	16
	2.3 Objective	18
3	Literature Review/Existing System	19
4	Basic System Architecture	23
	4.1 Outline of Proposed System	24
	4.2 Proposed System Block Diagram	27
5	Design	28
	5.1 Data Flow Diagram 5.2 Use Case Diagram 5.3 Class Diagram 5.4 Activity Diagram	
6	Implementation	34
	6.1 Algorithms	35
7	Testing	36
	7.1 Test Cases and Test Report	37
8	Project Screenshots	39
	8.1 Data visualization Screenshots	40
9	Conclusion & Future work	47

10	References	51

CHAPTER: 1

SYNOPSIS

1.Synopsis

1.1 Introduction



IMG.1

The **Land Price Prediction System** is a machine learning-based application developed to estimate the market value of land accurately. The system uses historical land data and various influencing factors such as location, area type, land type, amenities, risk levels, and connectivity to predict the price per square foot.

In many semi-urban and rural areas, land prices are usually decided through informal methods like broker suggestions or personal knowledge, which often leads to inaccurate pricing. This project aims to solve this problem by providing a data-driven and transparent solution for land price estimation.

The system is built using Python and the Flask framework, where the backend handles data processing and prediction, while the frontend provides an interactive user interface. A machine learning model, specifically Gradient Boosting, is used to achieve high prediction accuracy.

The application includes features such as land price prediction, cost calculation (including stamp duty and registration charges), investment suggestions (Buy, Hold, Wait, Caution), and comparison between different locations. This makes the system useful for buyers, sellers, investors, and government authorities.

Overall, the project demonstrates the practical use of machine learning in solving real-world problems in the real estate sector by improving decision-making and reducing uncertainty.

1.2 Literature Survey /Existing System



Img.2

The literature survey is an important part of any research project. It helps in understanding the existing systems, technologies, and research work related to the proposed project. In the field of real estate analysis and property price prediction, several researchers have applied machine learning and data analytics techniques to estimate property values and improve decision-making processes.

One of the significant research works titled **"Real Estate Price Prediction Using Machine Learning Techniques" (2020)** focuses on predicting property prices using machine learning algorithms such as Linear Regression, Decision Trees, and Random Forest. The study used historical property data including location, size of property, number of facilities, and infrastructure availability. The results showed that machine learning models can effectively analyze patterns in real estate data and generate accurate price predictions.

Another research study, **"Artificial Intelligence for Property Valuation and Investment Analysis" (2019)**, discusses the role of artificial intelligence in the real estate industry. The study explains how AI-based systems can analyze large datasets to determine property values and identify investment opportunities. By using AI techniques, the system can evaluate factors such as location advantages, accessibility, and environmental conditions to assist buyers and investors in making better decisions.

A research paper titled **"Property Price Estimation Using Data Mining Techniques" (2018)** explored the use of data mining methods to analyze real estate datasets and extract meaningful patterns related to property pricing. The study used clustering and regression techniques to predict property prices based on location and infrastructure development. The results indicated that data mining techniques can significantly improve the accuracy of property valuation models.

Another important study called **"Automated Property Valuation Using Machine Learning Models" (2021)** introduced an automated property valuation system that predicts property prices using machine learning algorithms. The system considered multiple parameters such as property size, location, neighborhood facilities, and accessibility. The study highlighted the importance of combining data analysis with intelligent algorithms to create reliable property evaluation systems.

In addition to price prediction, some research has also focused on calculating property-related taxes. A study titled **"Digital Systems for Property Tax and Stamp Duty Estimation" (2017)** proposed a system that calculates government taxes related to property transactions. This system helped users estimate additional costs such as stamp duty and registration charges before purchasing property.

Although many research works have focused on property price prediction or tax calculation, most existing systems do not provide intelligent recommendations about whether the land is suitable for purchase. The proposed system addresses this limitation by integrating **machine learning-based price prediction with an AI-based advisory module** that evaluates land quality and provides recommendations to the user.

By combining machine learning prediction models, tax estimation mechanisms, and AI-based decision support, the proposed system aims to provide a comprehensive solution for property buyers and investors.

1.3 Problem Statement

The **Land Price Prediction System** is designed to address the challenges of traditional land price estimation methods by providing a data-driven and automated solution. In many semi-urban and rural areas, land pricing lacks transparency and is often influenced by brokers, assumptions, or incomplete information. This system uses Machine Learning to deliver accurate predictions, insights, and better decision-making support.

Here's how the proposed system addresses the identified problems:

1. Lack of Transparent Price Estimation

Solution:
The system uses machine learning algorithms to analyze multiple factors such as location, area type, amenities, growth rate, and risk levels to predict land prices accurately.

Impact:
Provides fair and transparent pricing, reducing dependency on brokers and minimizing the chances of overpricing or underpricing.

2. Information Asymmetry Between Buyers and Sellers

Solution:
The system provides a centralized platform where users can input land details and get instant price predictions along with supporting insights.

Impact:
Helps buyers and sellers make informed decisions, reducing unfair advantages and improving trust in transactions.

3. Lack of Cost Awareness (Stamp Duty & Registration)

Solution:
The system automatically calculates stamp duty and registration charges based on area type and buyer gender, including applicable concessions.

Impact:
Users get a complete cost breakdown, increasing awareness and helping in better financial planning.

4. No Investment Decision Support

Solution:
The system provides investment signals such as **Buy Now, Hold, Wait, or Caution** based on growth rate, predicted price, and risk factors.

Impact:
Assists users in making smart investment decisions and reduces the risk of financial loss.

5. Absence of Risk Analysis

Solution:
The system considers risk factors like flood risk, crime level, and connectivity while predicting land prices.

Impact:
Provides a holistic view of the land value, helping users avoid risky investments.

6. Difficulty in Comparing Multiple Locations

Solution:
The system includes a comparison feature that allows users to analyze two locations side-by-side based on price, growth, and other attributes.

Impact:
Simplifies decision-making and helps users choose the best location according to their needs.

1.4 Objectives

1. Accurate Land Price Prediction

To develop a machine learning model that can predict land prices based on factors such as location, area type, amenities, and risk levels with high accuracy.

2. Comparison of Machine Learning Models

To analyze and compare different machine learning algorithms and select the best-performing model for reliable predictions.

3. Development of Web-Based Application

To design and implement a user-friendly web application using Flask that allows users to easily access prediction and analysis features.

4. Cost Calculation Automation

To calculate total land cost including stamp duty and registration charges automatically, considering area type and gender-based concessions.

5. Investment Decision Support

To provide intelligent suggestions such as **Buy Now, Hold, Wait, or Caution** based on growth rate, price comparison, and risk analysis.

6. Smart Location Recommendation

To suggest suitable locations to users based on their budget, preferences, and selected filters like distance, safety, and growth.

7. Location Comparison Feature

To allow users to compare two locations side-by-side based on price, features, and investment potential.

8. Transparency and Data-Driven Decision Makting

To provide a transparent system that helps users make informed decisions using data instead of assumptions.

1.5 Software requirement specification

1. Operating System

Windows 10 or higher will be used as the primary operating system for developing and running the application. The system can also run on other operating systems such as Linux or macOS.

2. Programming Language

Python will be used as the main programming language for implementing the machine learning model and backend functionality of the system.

3. Machine Learning Library

Scikit-learn library will be used to develop and train the machine learning model for land price prediction.

4. Data Processing Tools

Pandas and NumPy libraries will be used for data analysis, preprocessing, and handling the dataset used for training the prediction model.

5. Web Framework

Flask framework will be used to develop the backend web application and connect the machine learning model with the user interface.

6. Frontend Technologies

HTML, CSS, and JavaScript will be used to design the user interface and create interactive web pages for user input and result display.

7. Development Environment

Visual Studio Code (VS Code) or Jupyter Notebook will be used for coding, testing, and developing the machine learning model and application.

8. Database / Dataset

The system will use a dataset stored in CSV format containing information about land properties such as location, area, land type, road access, and government rates.

9. Web Browser

Google Chrome or any modern web browser will be used to run and test the web application.

1.6 Hardware requirement specification

The Land Price Prediction System is a software-centric project and does not require any external hardware components such as specialized land sensors or microcontrollers. The entire system operates using machine learning algorithms and web-based software. However, a computer system with sufficient processing capability is required to develop, train, and execute the AI model and the Flask application efficiently.

The following hardware components are required for the implementation of this project:

Component	Specification / Details	Purpose in Project
Computer/ Laptop	Intel i5 / Ryzen 5 processor or higher	Used for developing the system, training the Gradient Boosting and Random Forest models, and running the Flask backend.
RAM	Minimum 8 GB (16 GB recommended)	Required for running machine learning libraries (scikit-learn) and handling the 3,080-record dataset efficiently during processing. .
Storage	Minimum 100 GB available storage	Used to store the land dataset CSV, trained model files (model_clean.pkl), and system project files.
GPU (Optional)	NVIDIA GPU (if available)	Helps in accelerating potential deep learning future work and improves overall processing speed.
Input Devices	Keyboard and Mouse	Used for writing code in VS Code/Jupyter, interacting with the system, and providing user input for predictions.
Display Monitor	Standard monitor or laptop display	Used to visualize land price prediction results, analytical charts (Chart.js), and the interactive map dashboard (Leaflet.js)

Hardware Implementation Overview

- Software-Centric Design: The system relies on Python's computational power rather than external physical sensors to estimate land values.

- Scalability: Because it uses standard hardware, the system can be easily scaled to include more districts or larger datasets without significant infrastructure costs.
- Accessibility: The low hardware barrier ensures that the application can run on typical mid-range laptops used by students or real estate researchers

1.7 Methodology

The planning of work describes the step-by-step activities that will be carried out to successfully develop the proposed system. The development of the **AI Powered Land Price and Tax Prediction System with Smart Land Advisor** will be completed in several phases. Each phase focuses on a specific task required to build the system efficiently.

1. Problem Identification

The first step of the project involves identifying the problem in the real estate sector. Many people face difficulties in determining the actual land price and government taxes while purchasing land. Buyers often depend on brokers who may provide incorrect or misleading information. Therefore, the need for a smart system that predicts land prices and taxes and provides recommendations was identified.

2. Requirement Analysis

In this phase, the requirements of the system are analyzed. This includes identifying the system features such as land price prediction, tax estimation, infrastructure analysis, and AI-based recommendations. Hardware and software requirements required for system development are also determined during this stage.

3. Dataset Collection

The next step involves collecting or generating a dataset related to land properties. The dataset will include information such as location, land area, land type, road access, nearby facilities, government land rates, and other relevant factors. This dataset will be used for training the machine learning model.

4. Data Preprocessing

The collected dataset will be cleaned and prepared for machine learning analysis. This step includes removing duplicate records, handling missing values, and converting categorical data into numerical format so that the machine learning algorithm can process the data effectively.

5. Machine Learning Model Development

In this phase, machine learning algorithms will be applied to the dataset to train a prediction model. The model will learn patterns from the data and will be capable of predicting the estimated land price based on user input parameters such as location, area, and land type.

6. AI Recommendation Module Development

An AI-based advisory system will be developed to analyze additional parameters such as road connectivity, flood risk, transportation availability, and nearby facilities. Based on these factors, the system will provide suggestions indicating whether the land is suitable for purchase or not.

7. Web Application Development

A web-based interface will be developed where users can easily enter land details and obtain prediction results. Frontend technologies will be used to design the interface, while the backend system will connect the machine learning model with the web application.

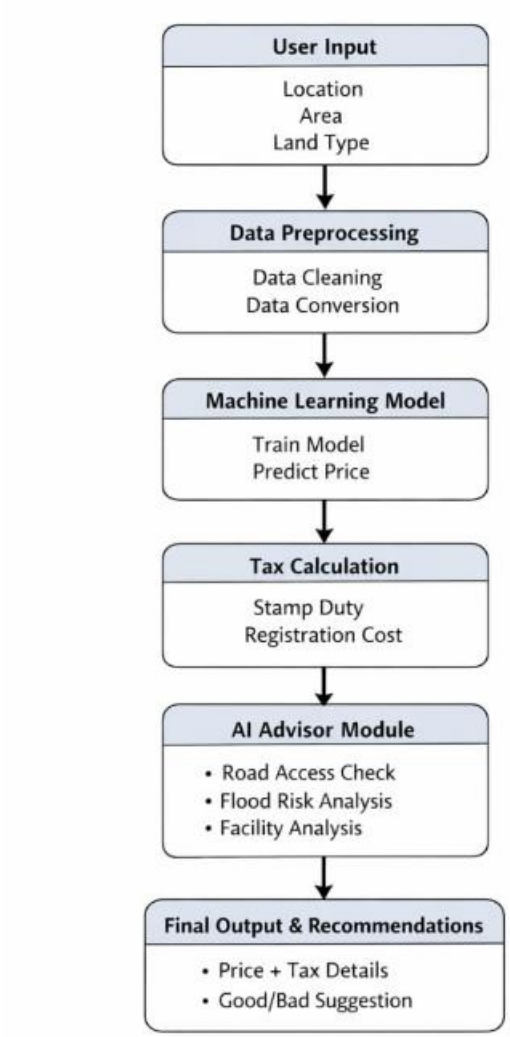
8. System Testing

After the development of the system, testing will be conducted to ensure that all modules work correctly. The accuracy of the prediction model and the functionality of the web application will be tested to verify that the system provides reliable results.

9. Implementation and Result Analysis

The final phase involves implementing the system and analyzing the results generated by the prediction model. The performance of the system will be evaluated based on prediction accuracy and user interaction.

1.8.Block diagram



1.9 References

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Mr. R. S. Todkar	Mr. R.S. Todkar	Mr.R.S. Kambale

CHAPTER: 2

INTRODUCTION

2.Introduction

2.1 Object Overview

The **Land Price Prediction System** is a machine learning-based application developed to estimate the market value of land accurately. The system uses historical land data and various influencing factors such as location, area type, land type, amenities, risk levels, and connectivity to predict the price per square foot.

In many semi-urban and rural areas, land prices are usually decided through informal methods like broker suggestions or personal knowledge, which often leads to inaccurate pricing. This project aims to solve this problem by providing a data-driven and transparent solution for land price estimation.

The system is built using Python and the Flask framework, where the backend handles data processing and prediction, while the frontend provides an interactive user interface. A machine learning model, specifically Gradient Boosting, is used to achieve high prediction accuracy.

The application includes features such as land price prediction, cost calculation (including stamp duty and registration charges), investment suggestions (Buy, Hold, Wait, Caution), and comparison between different locations. This makes the system useful for buyers, sellers, investors, and government authorities.

Overall, the project demonstrates the practical use of machine learning in solving real-world problems in the real estate sector by improving decision-making and reducing uncertainty.

2.2 Problem Statement

The **Land Price Prediction System** is designed to address the challenges of traditional land price estimation methods by providing a data-driven and automated solution. In many semi-urban and rural areas, land pricing lacks transparency and is often influenced by brokers, assumptions, or incomplete information. This system uses Machine Learning to deliver accurate predictions, insights, and better decision-making support.

Here's how the proposed system addresses the identified problems:

2.2.1. Lack of Transparent Price Estimation

Solution:
The system uses machine learning algorithms to analyze multiple factors such as location, area type, amenities, growth rate, and risk levels to predict land prices accurately.

Impact:
Provides fair and transparent pricing, reducing dependency on brokers and minimizing the chances of overpricing or underpricing.

2.2.2. Information Asymmetry Between Buyers and Sellers

Solution:
The system provides a centralized platform where users can input land details and get instant price predictions along with supporting insights.

Impact:
Helps buyers and sellers make informed decisions, reducing unfair advantages and improving trust in transactions.

2.2.3. Lack of Cost Awareness (Stamp Duty & Registration)

Solution:
The system automatically calculates stamp duty and registration charges based on area type and buyer gender, including applicable concessions.

Impact:
Users get a complete cost breakdown, increasing awareness and helping in better financial planning.

2.2.4. No Investment Decision Support

Solution:
The system provides investment signals such as Buy Now, Hold, Wait, or Caution based on growth rate, predicted price, and risk factors.

Impact:
Assists users in making smart investment decisions and reduces the risk of financial loss.

2.2.5. Absence of Risk Analysis

Solution:
The system considers risk factors like flood risk, crime level, and connectivity while predicting land prices.

Impact:
Provides a holistic view of the land value, helping users avoid risky investments.

2.2.6. Difficulty in Comparing Multiple Locations

Solution:
The system includes a comparison feature that allows users to analyze two locations side-by-side based on price, growth, and other attributes.

Impact:
Simplifies decision-making and helps users choose the best location according to their needs.

2.3 Objectives

2.3.1. Accurate Land Price Prediction

To develop a machine learning model that can predict land prices based on factors such as location, area type, amenities, and risk levels with high accuracy.

2.3.2. Comparison of Machine Learning Models

To analyze and compare different machine learning algorithms and select the best-performing model for reliable predictions.

2.3.3. Development of Web-Based Application

To design and implement a user-friendly web application using Flask that allows users to easily access prediction and analysis features.

2.3.4. Cost Calculation Automation

To calculate total land cost including stamp duty and registration charges automatically, considering area type and gender-based concessions.

2.3.5. Investment Decision Support

To provide intelligent suggestions such as **Buy Now, Hold, Wait, or Caution** based on growth rate, price comparison, and risk analysis.

2.3.6. Smart Location Recommendation

To suggest suitable locations to users based on their budget, preferences, and selected filters like distance, safety, and growth.

2.3.7. Location Comparison Feature

To allow users to compare two locations side-by-side based on price, features, and investment potential.

2.3.8. Transparency and Data-Driven Decision Making

To provide a transparent system that helps users make informed decisions using data instead of assumptions.

CHAPTER: 3

LITERATURE SURVEY

3.Literature survey / Existing system

The study of land and property price prediction has gained significant attention in recent years due to the rapid growth of the real estate sector and the need for accurate valuation methods. Various researchers and organizations have explored different approaches ranging from traditional statistical techniques to advanced machine learning models.

3.1 Traditional Approaches

In earlier years, land price prediction was primarily based on statistical models such as linear regression and hedonic pricing methods. The hedonic pricing model considers the price of land as a function of its characteristics, including location, size, accessibility, and available amenities. These models are simple, interpretable, and computationally efficient.

However, traditional methods have several limitations. They assume a linear relationship between independent variables and the target variable, which is not always true in real-world scenarios. Land prices are influenced by complex interactions between multiple factors such as infrastructure development, environmental risks, and market trends. As a result, traditional models often fail to provide highly accurate predictions, especially in diverse and non-linear datasets.

3.2 Machine Learning-Based Approaches

With the advancement of technology, machine learning techniques have been widely adopted for real estate price prediction. Algorithms such as Decision Trees, Random Forest, Support Vector Machines, and Gradient Boosting have shown promising results.

- **Decision Trees** are easy to understand and can capture non-linear relationships, but they may suffer from overfitting.
- **Random Forest**, an ensemble of decision trees, improves accuracy and reduces overfitting by combining multiple trees.
- **Support Vector Machines (SVM)** are effective for high-dimensional data but require careful parameter tuning.
- **Gradient Boosting** builds models sequentially, where each new model corrects the errors of the previous ones, leading to high prediction accuracy.

Among these, Gradient Boosting has been widely recognized as one of the most effective algorithms for regression problems due to its ability to handle complex patterns and reduce prediction errors.

3.3 Existing Real Estate Systems

Several online platforms provide property price estimation services. However, these systems have certain limitations:

- They mainly focus on urban and metropolitan areas, with very limited data available for rural or semi-urban regions.
- They provide only approximate price ranges without explaining the influencing factors.
- They do not include important aspects such as environmental risks, crime levels, or future growth potential.
- They lack transparency in their prediction methods, making it difficult for users to trust the results.

Government platforms, such as stamp duty and registration portals, provide official valuation guidelines but do not offer real-time market price predictions or investment analysis.

3.4 Research Gap

Despite the availability of various models and systems, there is still a significant gap in providing a comprehensive land price prediction system for semi-urban and rural areas. Most existing research focuses on metropolitan cities and does not consider region-specific factors.

Additionally, very few systems integrate multiple features such as:

- Risk analysis (flood risk, crime level)
- Growth rate and investment potential
- Cost calculation including stamp duty and registration
- User-friendly comparison and recommendation tools

There is also a lack of systems that provide both prediction accuracy and interpretability, along with practical usability for common users.

3.5 Proposed System Contribution

The proposed **Land Price Prediction System** aims to overcome the limitations of existing approaches by integrating machine learning with domain-specific knowledge. The system uses advanced algorithms, particularly Gradient Boosting, to achieve high accuracy in predicting land prices.

It incorporates multiple features such as location, amenities, connectivity, risk factors, and growth rate to provide a more realistic estimation. In addition, the system offers:

- Automatic cost calculation including stamp duty and registration charges

- Investment decision support (Buy, Hold, Wait, Caution)
- Smart recommendation of locations based on user preferences
- Comparison of multiple locations for better decision-making

By combining these features into a single platform, the system provides a comprehensive and practical solution for land price prediction.

CHAPTER: 4

BASIC SYSTEM ARCHITECTURE

4.Basic System Architecture

4.1 Outline of Proposed System

The **Land Price Prediction System** is designed using a structured and modular architecture that ensures efficient data processing, accurate prediction, and smooth user interaction. The system follows a **three-tier architecture**, which separates the application into frontend, backend, and machine learning components. This design improves scalability, maintainability, and performance.

4.1.1 Overview of System Architecture

The system architecture consists of three main layers:

1. Presentation Layer (Frontend)
2. Application Layer (Backend)
3. Data & Machine Learning Layer

Each layer performs a specific function and communicates with the others to complete the overall process of land price prediction.

4.2 Presentation Layer (Frontend)

The presentation layer is the user interface of the system. It is developed using **HTML, CSS, and JavaScript**. This layer allows users to interact with the system by entering inputs and viewing results.

Functions of Frontend:

- Provides forms for user input (location, area, gender, budget, etc.)
- Displays predicted land price and cost breakdown
- Shows charts, graphs, and comparison results
- Provides navigation between different modules

The frontend is designed to be simple and user-friendly so that even non-technical users can easily operate the system.

4.3 Application Layer (Backend)

The backend is developed using **Python and Flask framework**. It acts as a bridge between the user interface and the machine learning model.

Functions of Backend:

- Receives input data from the user
- Validates and processes the input
- Performs feature engineering
- Sends processed data to the machine learning model
- Receives prediction results
- Calculates total cost (land price, stamp duty, registration)
- Generates investment signals
- Sends final output back to the frontend

Flask handles routing, request processing, and communication between different components of the system.

4.4 Data and Machine Learning Layer

This layer is the core of the system where data processing and prediction take place.

4.4.1.Dataset

The dataset contains land-related information such as:

- Location (village, taluka)
- Area type (urban, rural, etc.)
- Land type
- Amenities (school, hospital, market access)
- Risk factors (flood risk, crime level)
- Distance from city and highway
- Rate per square foot (target variable)

4.4.2.Feature Engineering

New features are created to improve model performance:

- Location Score
- Connectivity
- City Proximity
- Risk Score

4.4.3.Machine Learning Model

The system uses **Gradient Boosting Regressor** for prediction.
This model was selected after comparing multiple algorithms due to its high accuracy and ability to handle complex data relationships.

4.4.4.Model Storage

The trained model is saved using **Joblib**, which allows quick loading during prediction without retraining.

4.5 Working of the System

The working process of the system can be explained in the following steps:

1. User opens the web application and logs in
2. User enters required inputs (location, area, gender, etc.)
3. Backend receives and validates the input
4. Feature engineering is applied to the input data
5. Processed data is sent to the machine learning model
6. Model predicts the land price
7. Backend calculates total cost including stamp duty and registration
8. Investment signal is generated
9. Final results are displayed to the user

4.6 System Modules

The system is divided into three main modules:

4.6.1. Land Price Prediction Module

- Predicts price per square foot
- Shows cost breakdown
- Displays investment signal

4.6.2. Smart Recommendation Module

- Suggests best locations based on budget and filters
- Displays results with map visualization

4.6.3. Location Comparison Module

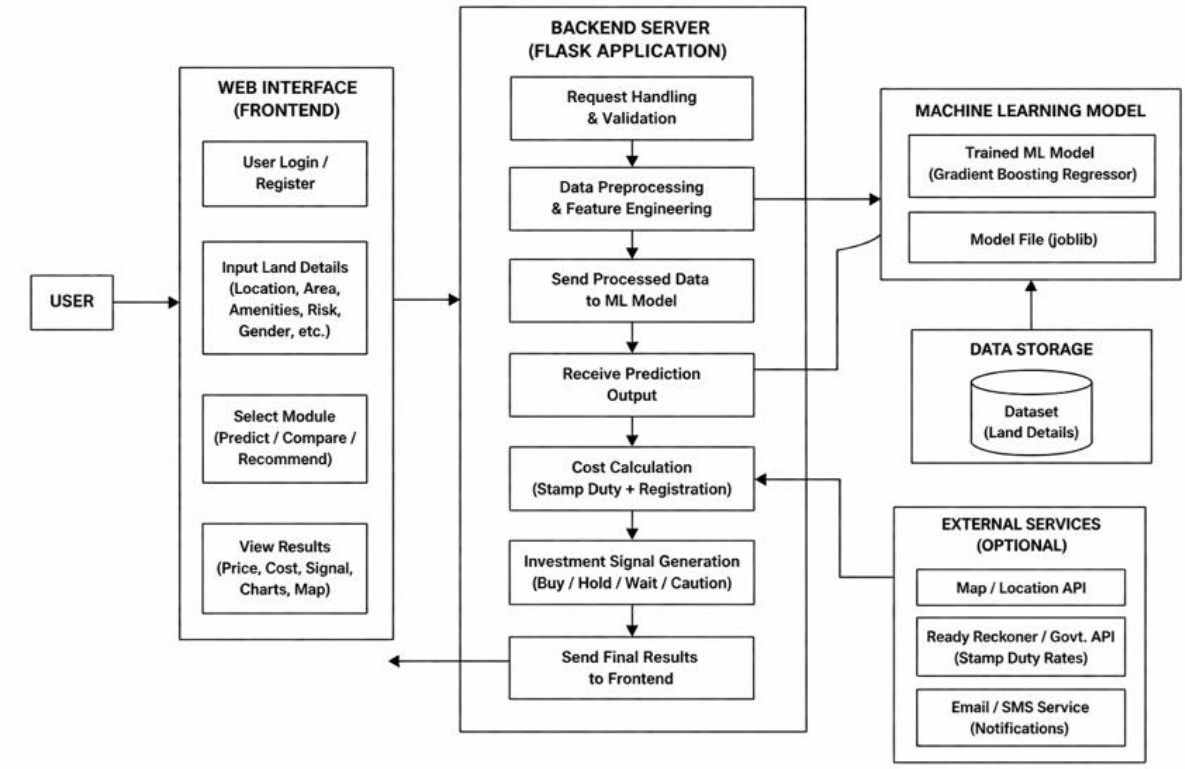
- Compares two locations

- Shows differences in price, growth, and features

4.7 Advantages of Architecture

- Clear separation of components
- Easy to maintain and update
- Scalable for future enhancements
- Efficient communication between layers
- Supports real-time prediction

Proposed system block diagram



CHAPTER: 5

DESIGN

5. Design

5.1 Data Flow Diagram

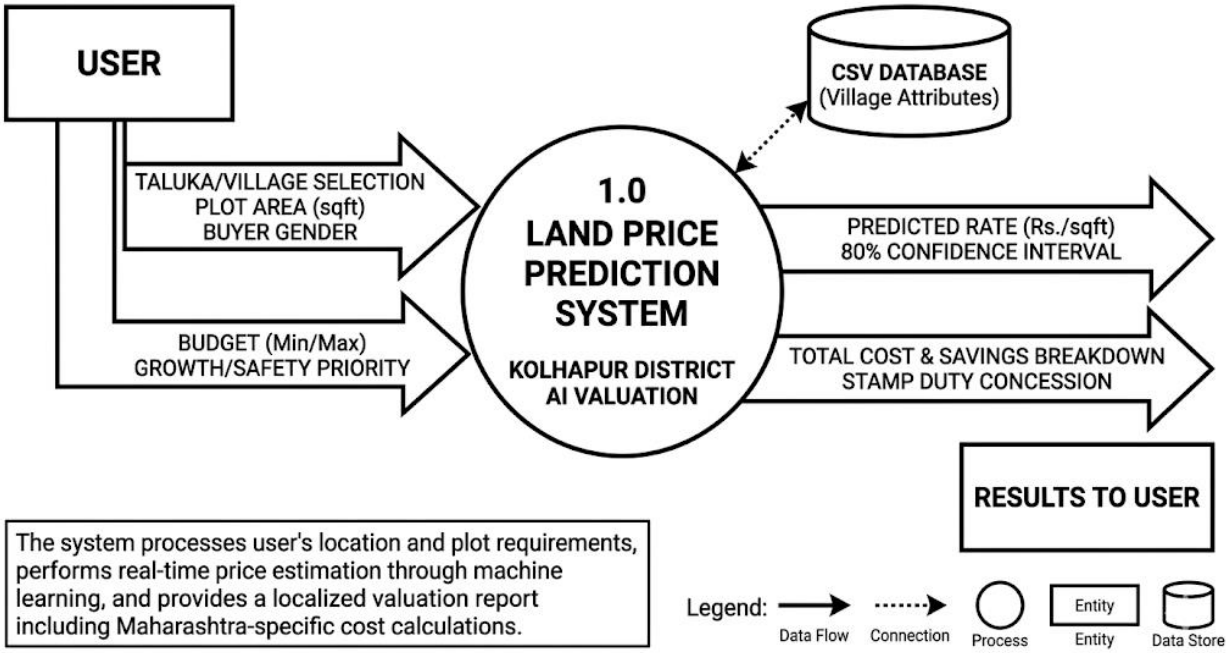


Fig. 5.1 DFD Level 0

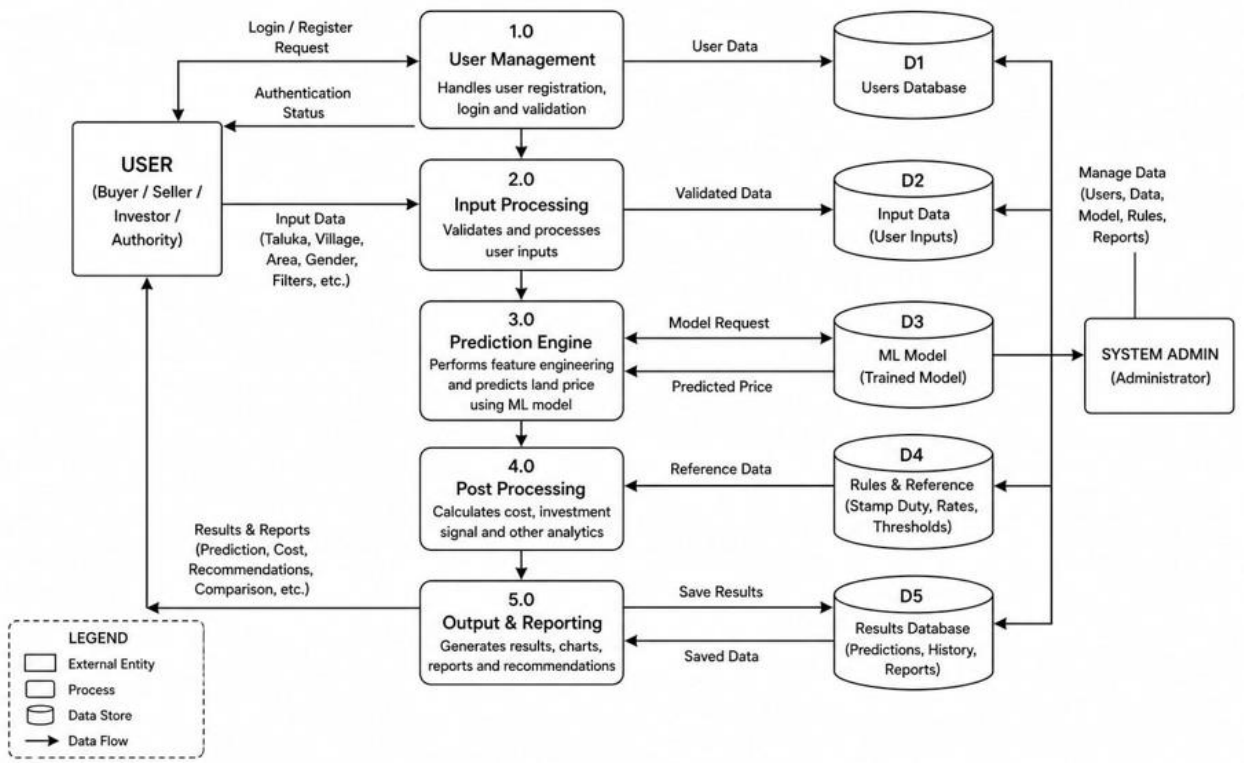
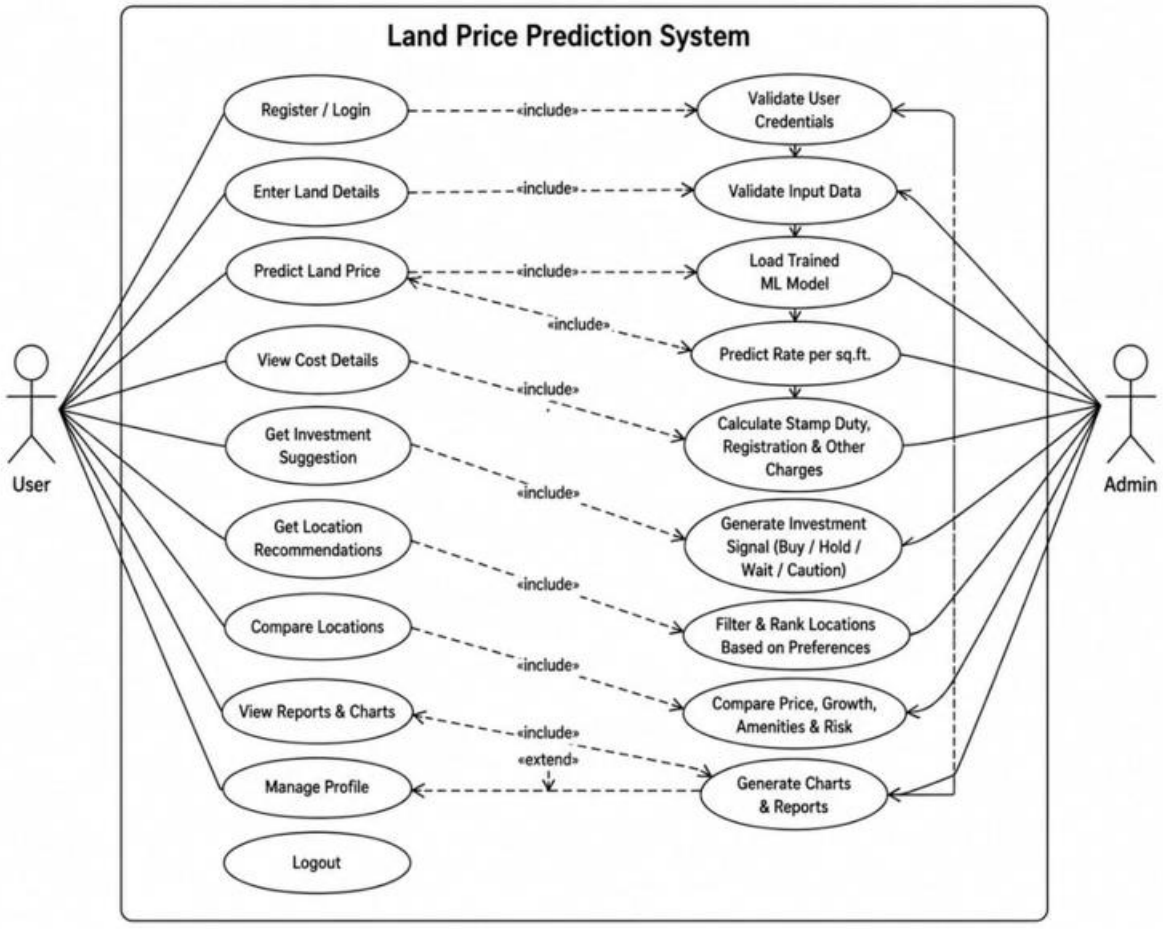
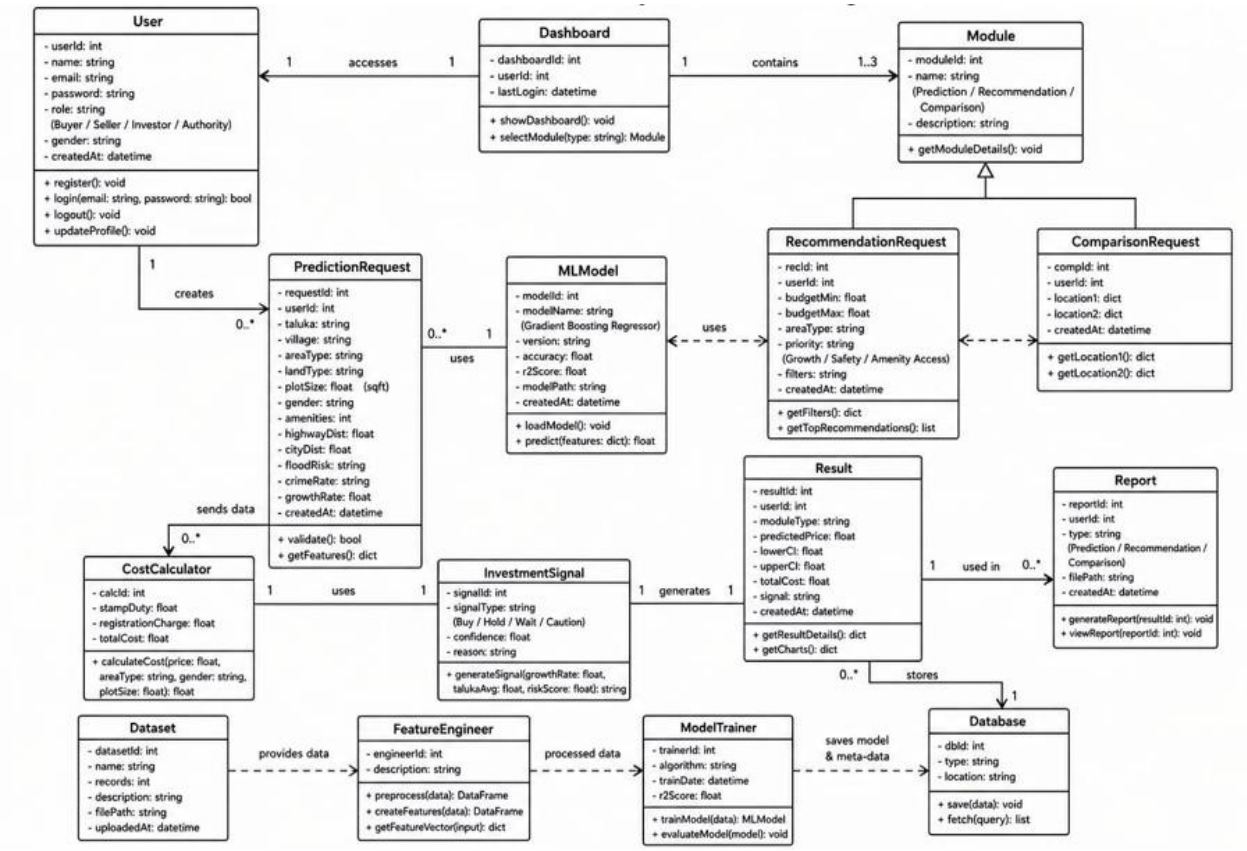


Fig. 5.1 DFD Level 1

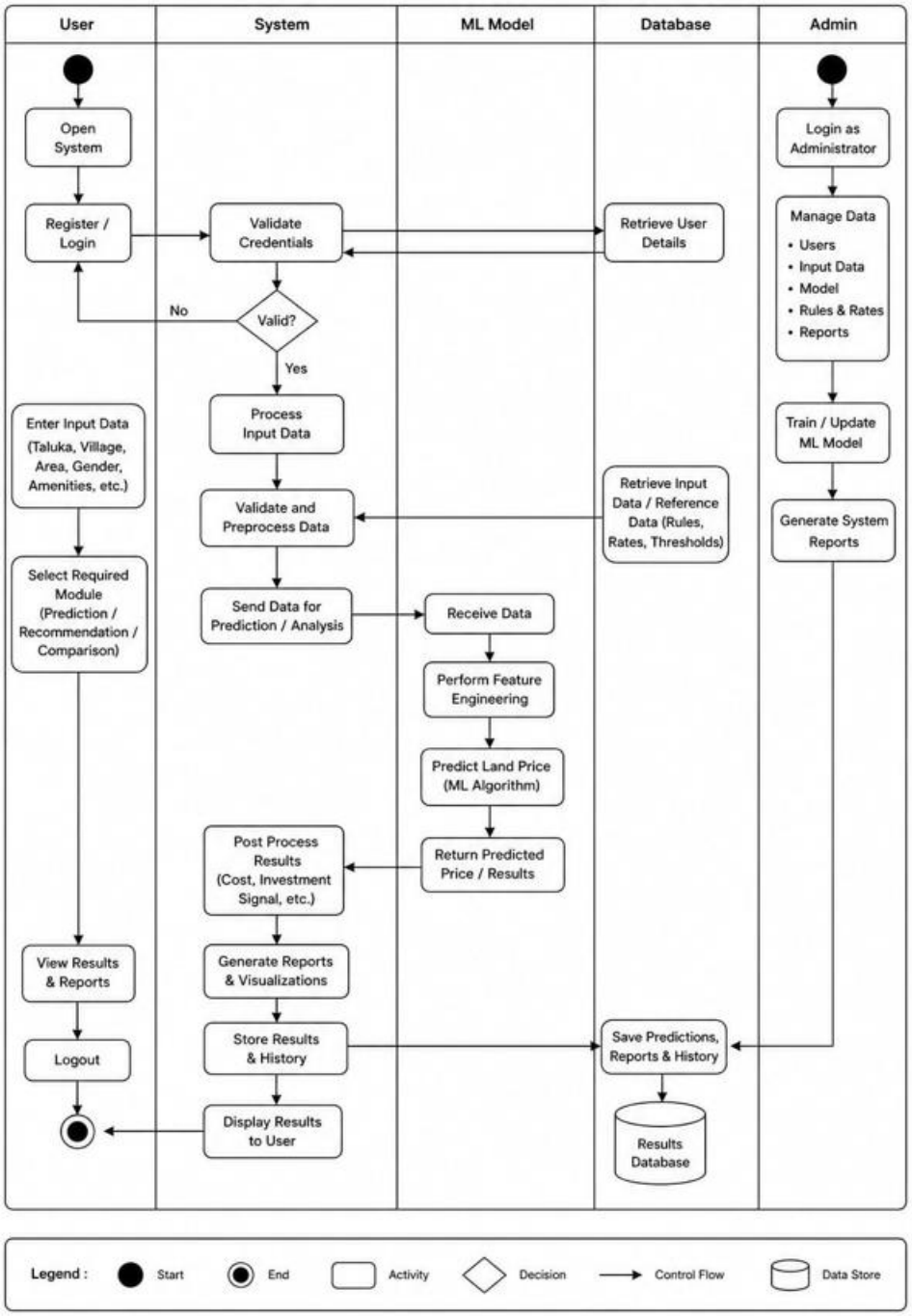
5.2 Use Case Diagram



class5.3 Class Diagram



5.4 Activity Diagram

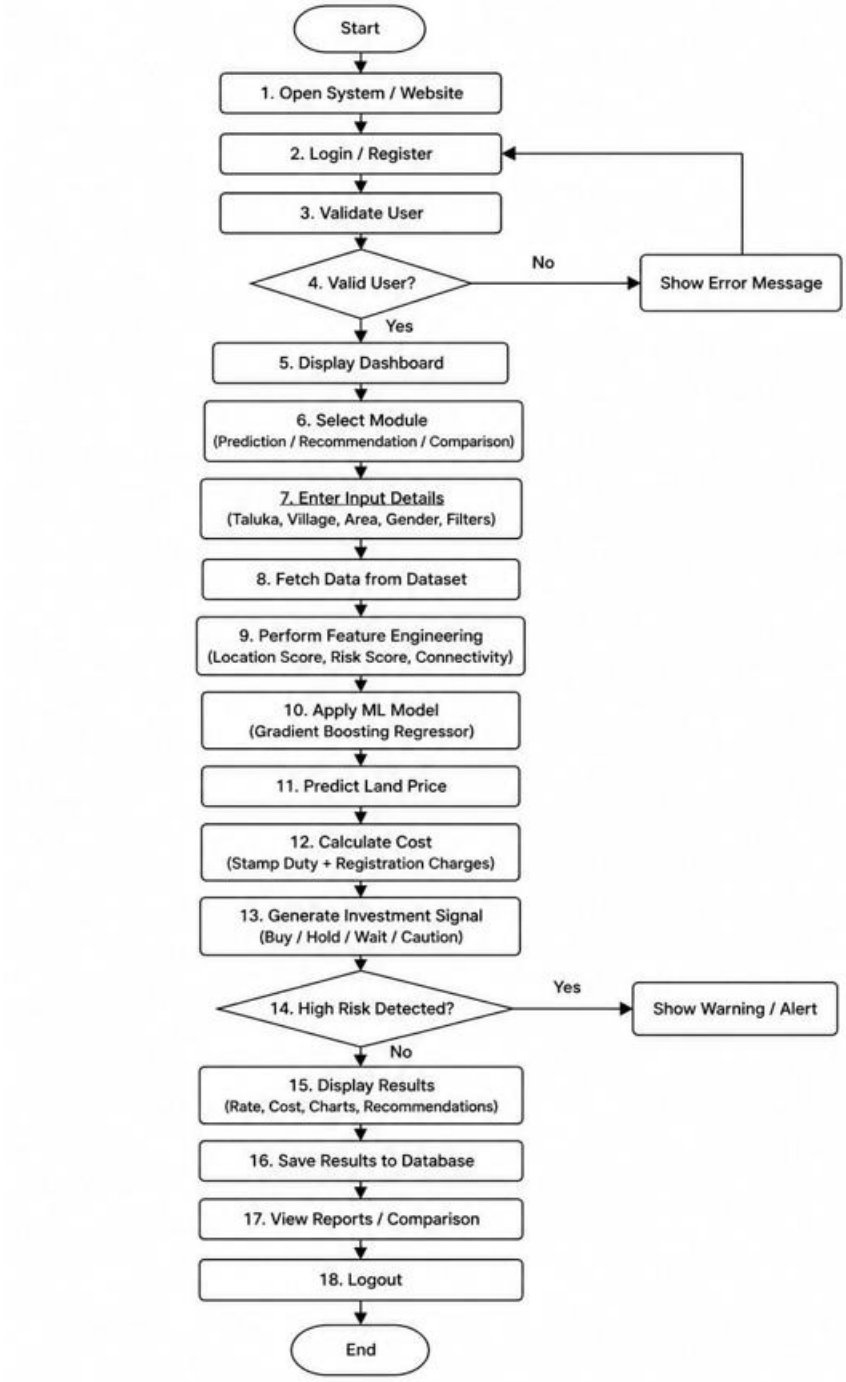


CHAPTER: 6

IMPLEMENTATION

6. Implementation

6.1 Algorithm



CHAPTER: 7

TEST CASES

Test Case ID	CRAI_01	Test Case Description	The Land Price Prediction System was tested using different test cases to ensure proper functionality, accuracy, and user satisfaction. Each module of the system was checked with valid and invalid inputs.		
Created By	Om A. Karande Parth K. Patil Prathamesh P. Furade Neha N. Patil Sanjyot S. patil	Reviewed By	Mr.Rajwardhan S. Todkar	Version	1.0
QA Tester's Log	Initial testing complete in version 1.0				
Tester's Name	Om Karande, Parth Patil, Prathamesh Furade, Neha Patil, Sanjyot Patil	Date Tested	07/04/2026	Test Case (Pass/Fail)	Pass

S #	Prerequisites:
1	Python Installed
2	Required Libraries Installed
3	Land Dataset
4	VS Code / Jupyter Notebook
5	Web Browser and Computer System

S #	Test Data
1	User Login Details
2	Location Name and Area
3	Land Type Details
4	Budget and Buyer Gender
5	Amenities and Risk Data

Test Scenario: Verify that the system successfully predicts accurate land prices and displays complete results when the user enters valid land details.

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not executed / Suspended
1	Open system and login	Dashboard should open successfully	As Expected	Pass
2	Taluka: Karveer, City: Rajarampuri, Area: 1500 sqft, Gender: Male	Rate: ~Rs.6,500/sqft, Stamp Duty: 6%, Signal: BUY NOW or HOLD	As Expected	Pass
3	Same location, Gender: Female	Stamp Duty: 5% (1% less than Male, saving ~Rs.9,900 on 1500 sqft)	As Expected	Pass
4	Area: 0 (invalid input)	Form validation error, no prediction	As Expected	Pass

5	No city selected (empty dropdown)	Form submission blocked	As Expected	Pass
6	Budget: Rs.50L, Area: 1000 sqft, Priority: Growth, No filters	Top 5 locations ranked by growth rate, map with 5 pins	As Expected	Pass
7	Max Crime: Low only, Max Distance: 10 km	Only low-crime villages within 10 km of Kolhapur	As Expected	Pass
8	Same comparison, Gender: Female	Stamp changes to 5% and 2% respectively, totals reduced	As Expected	Pass
9	Same location for both (Rajarampuri vs Rajarampuri	Equal scores, same rates, winner defaults to Location 1	As Expected	Pass
10	Model Report page accessed	4-model comparison table, feature importance chart, holdout results for 5 villages	As Expected	Pass

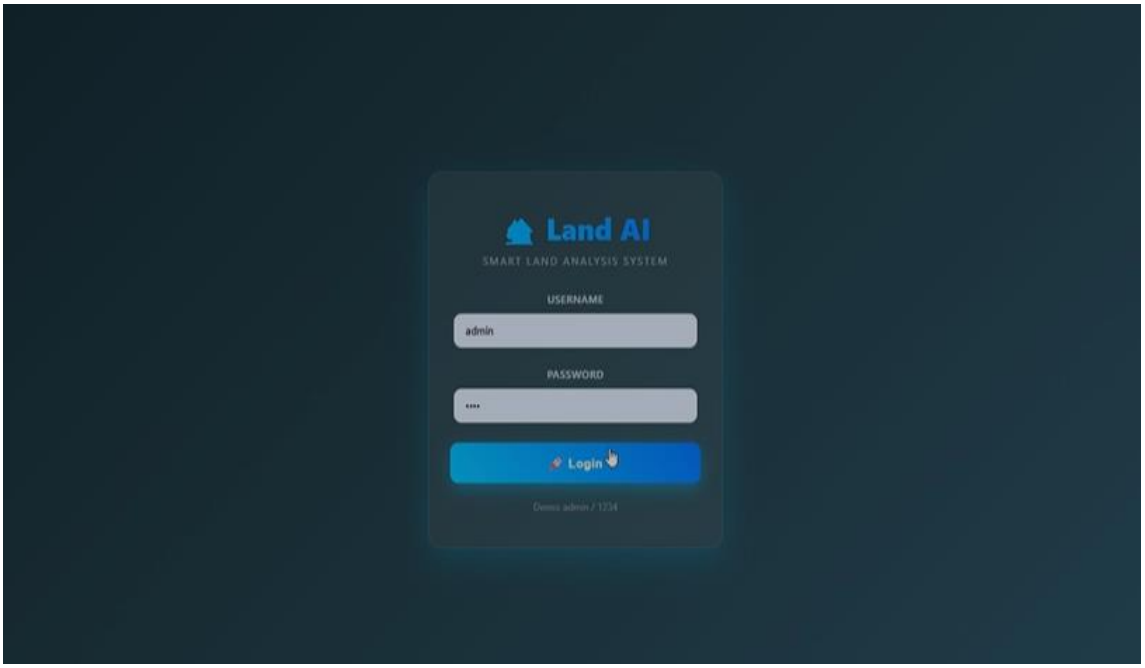
CHAPTER: 8

PROJECT SCREENSHOTS

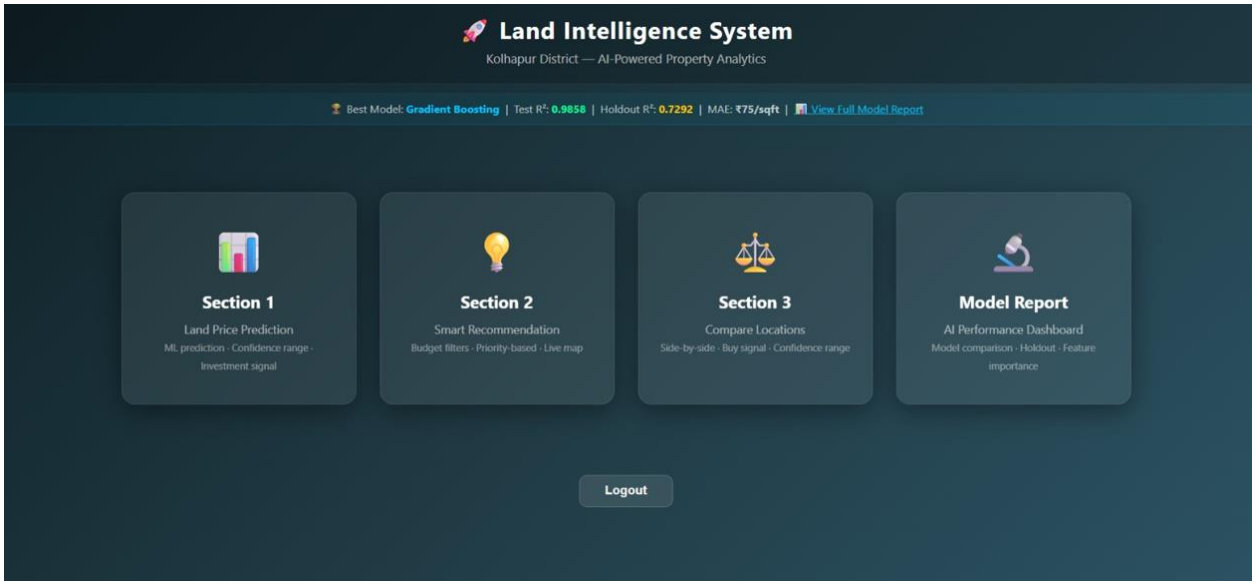
8. Project Screenshots

8.1 Data visualization Screenshots

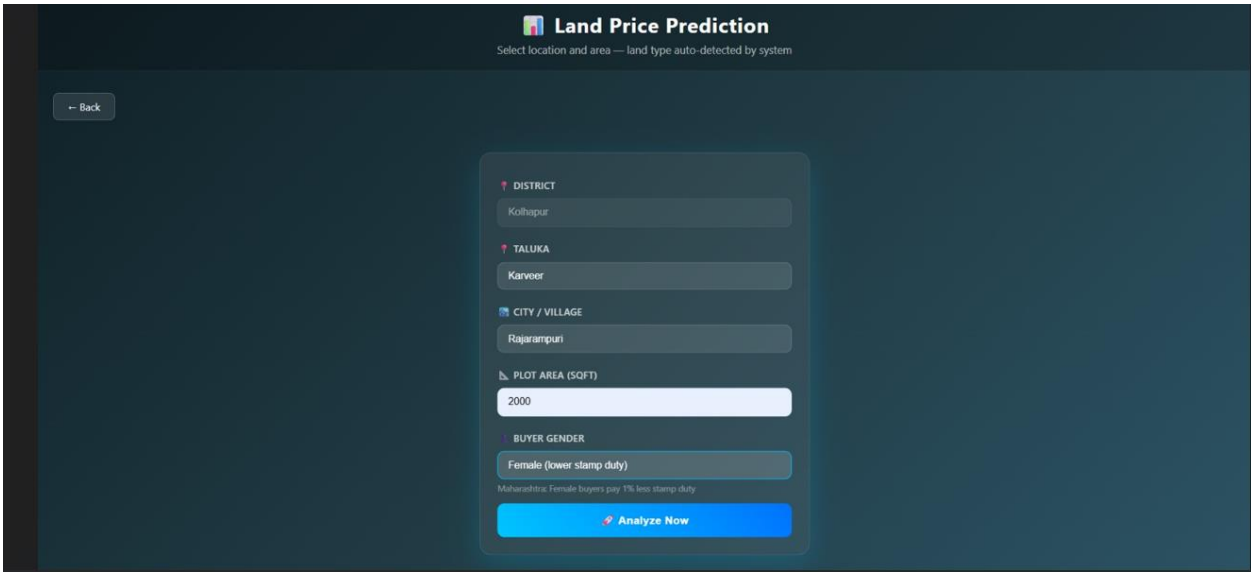
1. Login Page



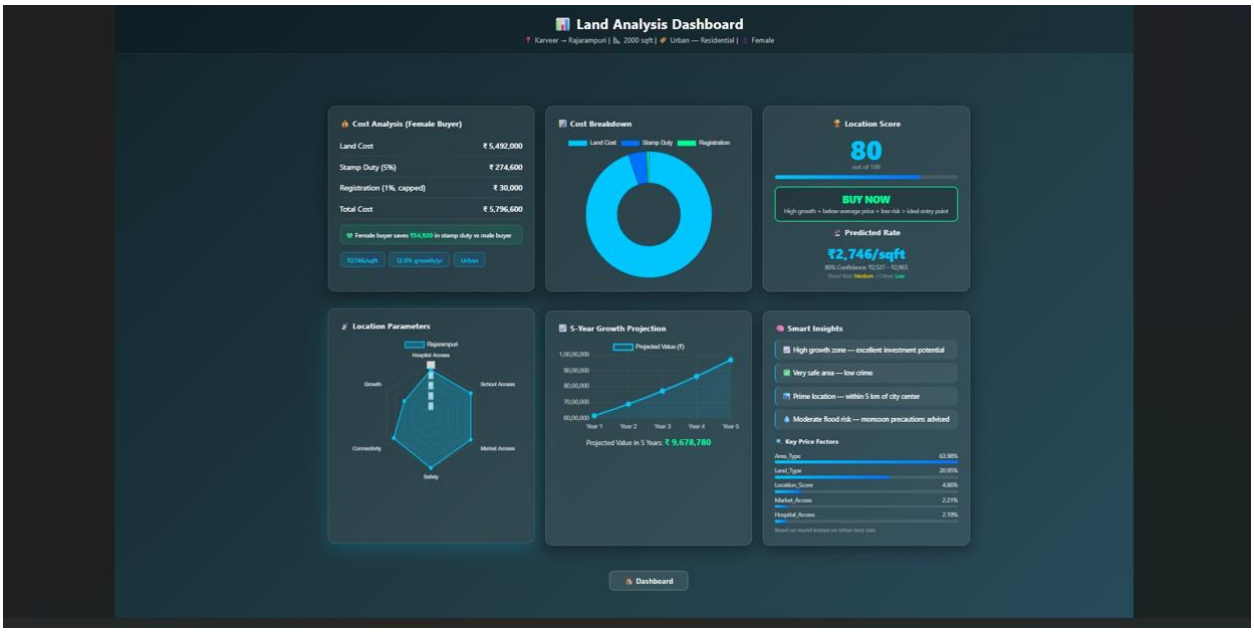
2.Land Intelligence System



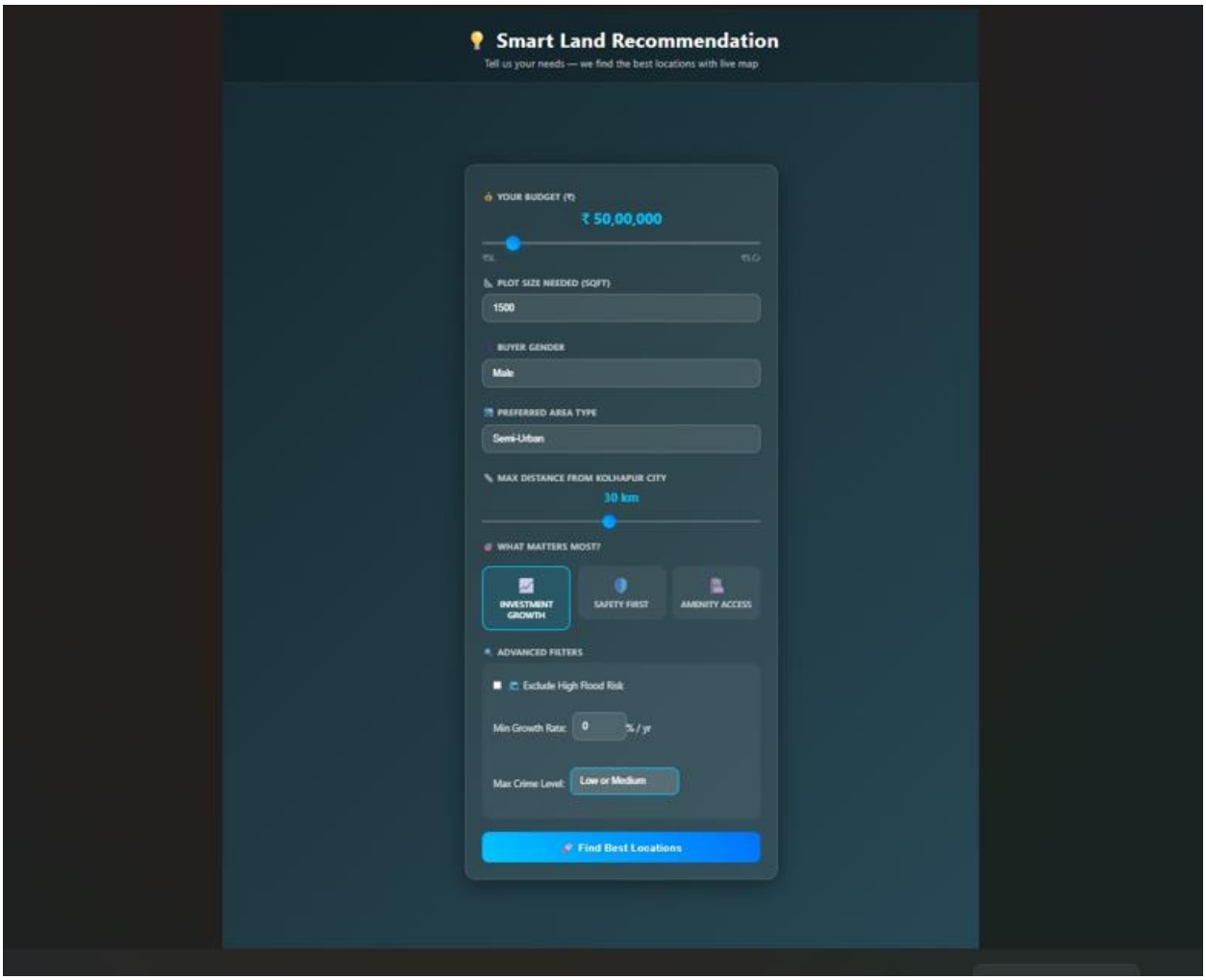
3.Land Price Prediction



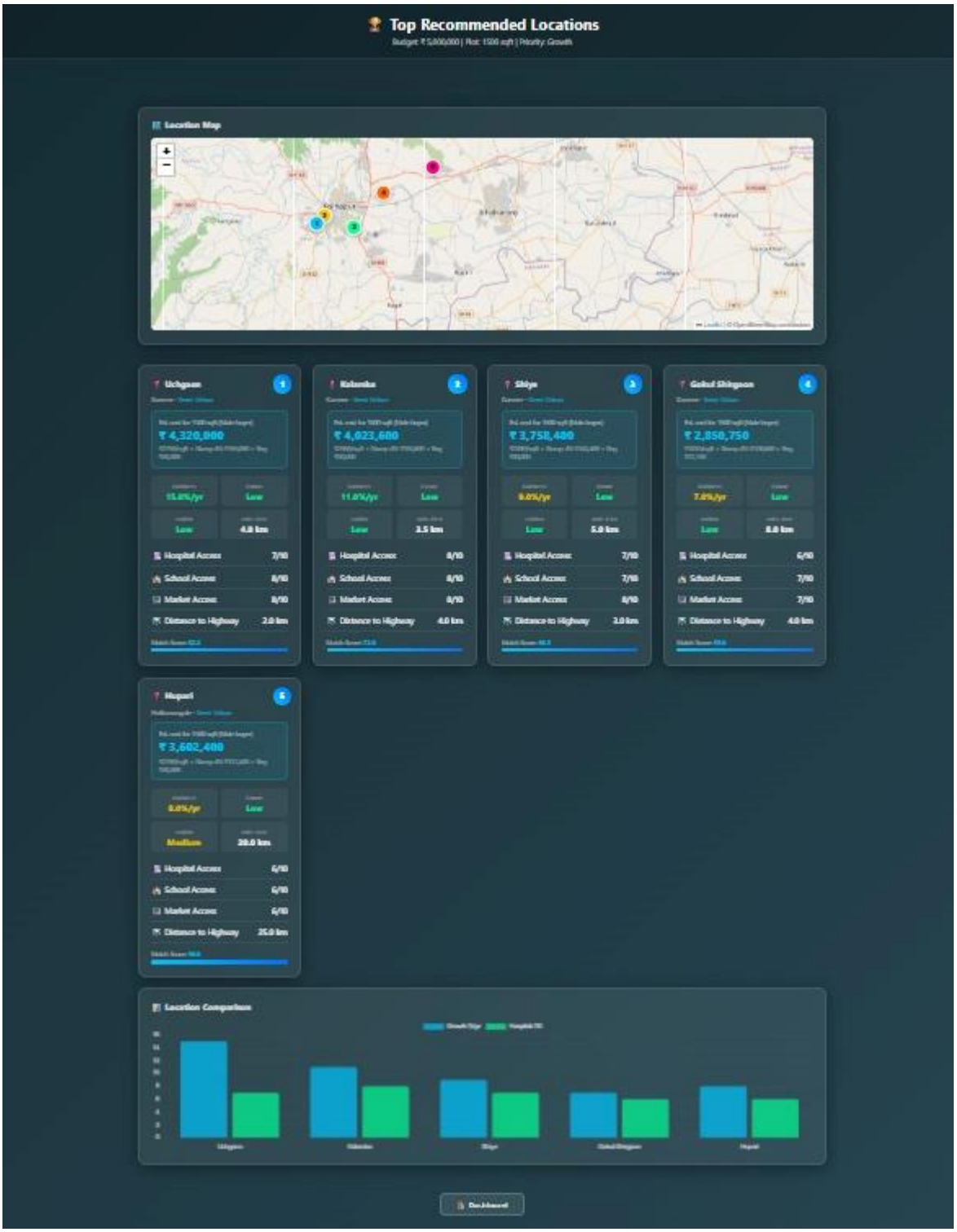
4. Dashboard



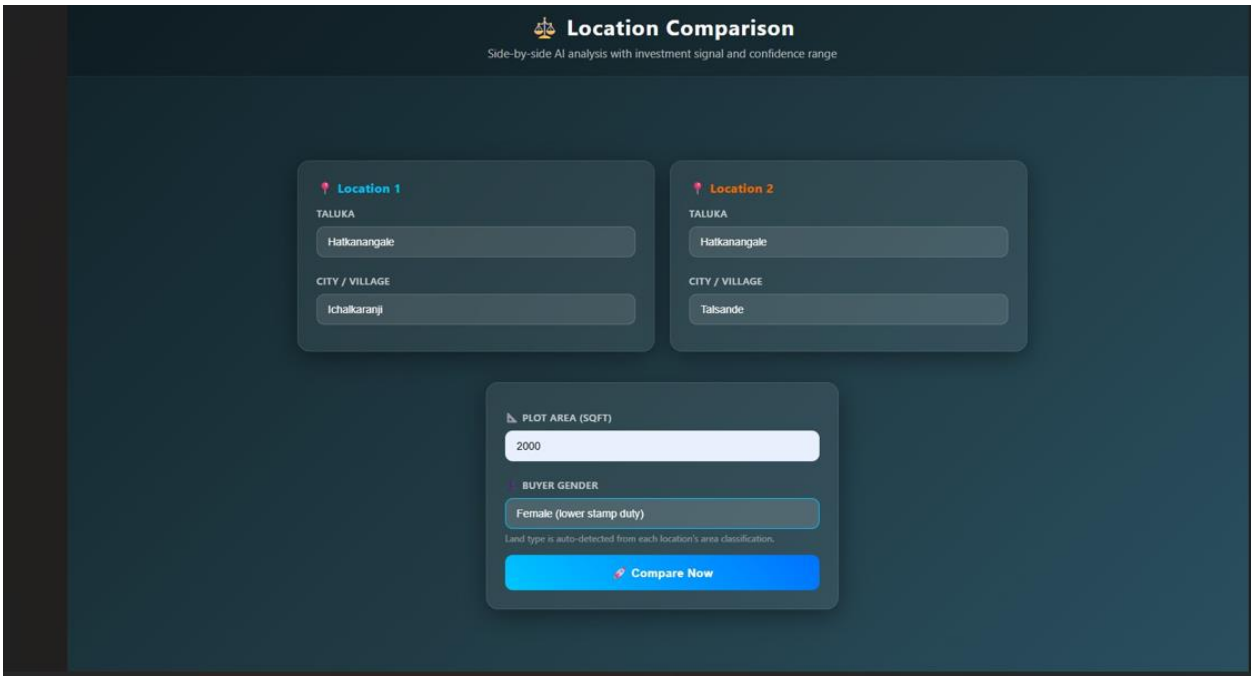
5.Smart Land Recommendation



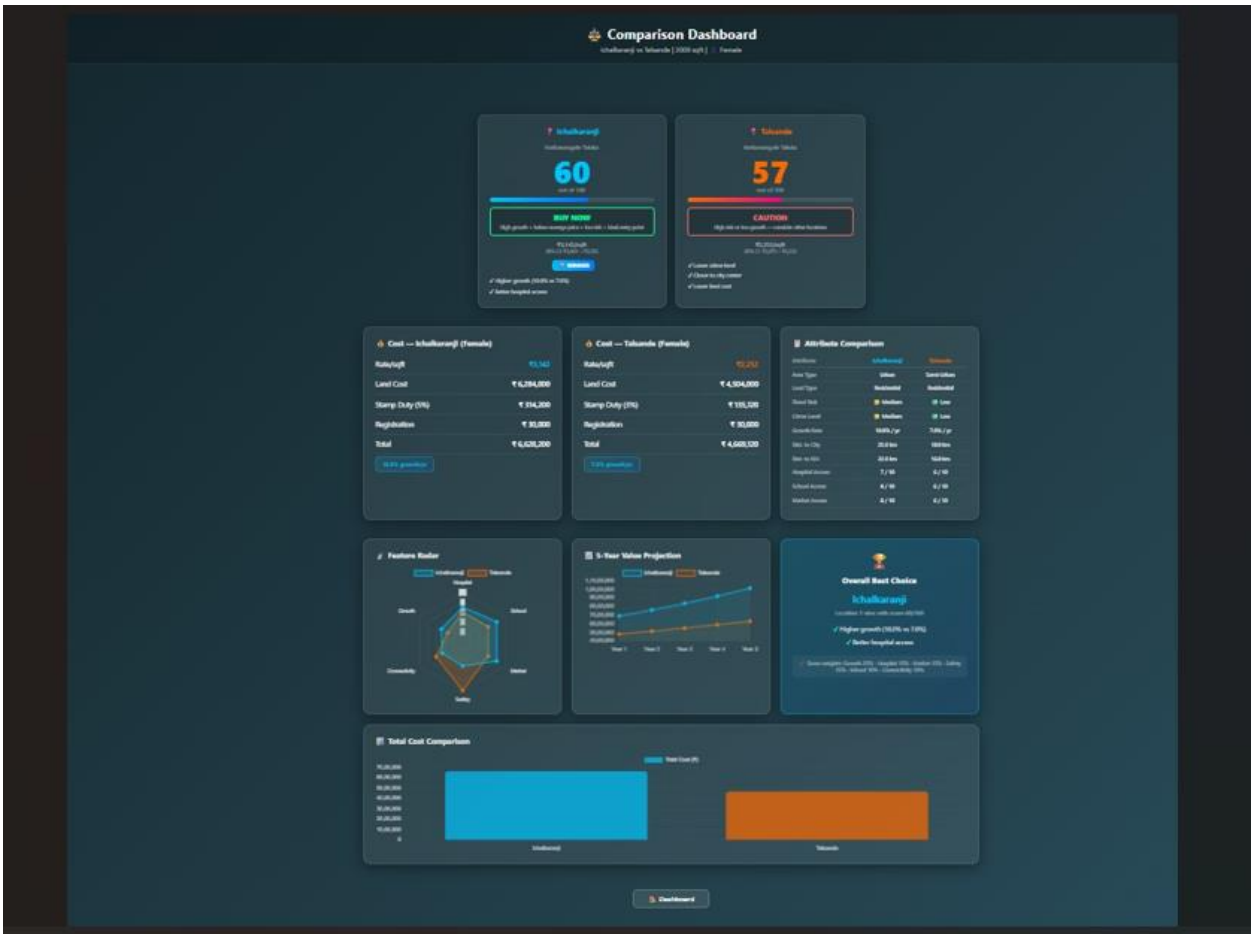
6.Top Recommended Locations



7. Location Comparison



8.Comparison Dashboard



9.AI Model Performance Report



CHAPTER: 9

CONCLUSION & FUTURE WORK

9.Conclusion & Future Work

9.1 Conclusion

The Land Price Prediction System successfully demonstrates the practical application of Machine Learning to a real-world socio-economic problem—providing transparent and data-driven land valuation for the Kolhapur district of Maharashtra. The system achieved a high level of technical accuracy, specifically a **Test R-squared of 0.9858** using the Gradient Boosting Regression algorithm. Key takeaways from the project include:

- **Model Generalization**: The system achieved a Holdout R-squared of 0.7292 on five completely unseen villages, proving that the model can generalize beyond its initial training data.

- **Domain Integration**: Unlike standard ML models, this system incorporates specific Maharashtra government rules, such as the 1% stamp duty concession for female buyers and the registration fee cap of Rs. 30,000.

- **Actionable Intelligence**: By providing investment signals like BUY NOW, HOLD, WAIT, or CAUTION, the system translates raw data into clear decision-making support for users.

- **Risk Quantification**: The inclusion of an 80% confidence interval on each prediction allows users to understand the mathematical uncertainty of the estimate, a feature often missing from commercial real estate tools.

- **Academic Rigor**: The project successfully covers essential AIML competencies, including ensemble methods, feature engineering, cross-validation, and full-stack web API design.

9.2.Future Work

While the current system provides a robust foundation, several areas have been identified for future enhancement to increase its real-world utility:

9.2.1.Real Data Integration: The system can be improved by connecting to the Maharashtra IGR's Ready Reckoner API and web-scraping commercial portals (like 99acres or MagicBricks) to replace synthetic data with live transaction records.

9.2.2.Deep Learning Models: Future iterations could experiment with Feedforward Neural Networks using TensorFlow or Keras to compare their performance against Gradient Boosting for complex, non-linear interactions.

9.2.3.Mobile Application Development: Developing a dedicated mobile app (using Flutter or React Native) would allow real estate agents and government surveyors to use the tool for offline-capable field predictions.

9.2.4.Advanced Forecasting: Incorporating time-series models like LSTM or Prophet would enable the system to predict price trajectories 3–10 years into the future with seasonal adjustments.

9.2.5.Sentiment Analysis: Integrating NLP-based analysis of local Marathi news could help the model factor in neighborhood development plans and infrastructure projects into its price predictions.

9.3.6.Geographic Expansion: The dataset could be extended to cover all districts in the Konkan and Marathwada regions, eventually creating a pan-Maharashtra valuation system.

9.2.7.Government Collaboration: Proposing integration with the SARATHI portal would make the tool officially useful for legal transactions and automatic stamp duty calculations linked to live rates

CHAPTER: 10

REFERENCES

10. References

1] Research Papers:

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2.Maharashtra Inspector General of Registration (IGR). (2024). Ready Reckoner Rates 2024–25. Government of Maharashtra. Available at: https://igrmaharashtra.gov.in

3.Breiman, L. (2001). Random Forests. Machine Learning, 45(1), 5–32. https://doi.org/10.1023/A:1010933404324

2] Online Resources:

1. Scikit-learn Official Documentation:

https://scikit-learn.org

2.Flask Official Documentation:

https://flask.palletsprojects.com

3.Maharashtra IGR Official Website:

https://igrmaharashtra.gov.in